

Goal. Trig substitution: making integrals simpler (or at least doable)

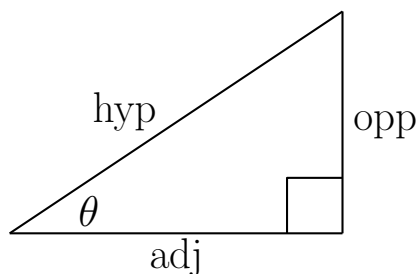
1. Evaluate the integral by the following steps.

$$\int \frac{1}{x^4 \sqrt{x^2 - 4}} dx$$

- (a) First, substitute $x = 2 \sec(\theta)$ (this is a *trig substitution*). Convert the integral completely into the variable θ . (Hint: If $x = 2 \sec(\theta)$ then what does dx equal?)

- (b) Use the trig identity $\tan^2(\theta) + 1 = \sec^2(\theta)$ to simplify the denominator by making the part under the square root into a perfect square. (This kind of simplification is the goal of trig substitution.) Then evaluate the integral.

- (c) Convert back to the variable x by using our trig substitution $\sec(\theta) = \frac{x}{2}$ and the definition of the trig functions in terms of a right triangle.



$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad \sec \theta =$$

$$\cos \theta = \quad \csc \theta =$$

$$\tan \theta = \quad \cot \theta =$$

2. Evaluate the integral by the following steps.

$$\int \frac{x}{(x^2 - 6x + 5)^{5/2}} dx$$

3. Complete the square in the denominator.

$$x^2 - 6x + 5 = (x^2 - 6x + \underline{\hspace{1cm}}) + 5 - \underline{\hspace{1cm}} = (x - 3)^2 - \underline{\hspace{1cm}}$$

4. Now make the substitution $u = x - 3$ to get the integrand into a form where we can use trig substitution. Convert the integral completely into the variable u .

5. Now set u equal to something involving a trig function of θ that will turn the expression in the denominator into a perfect square. Convert the integral completely into the variable θ . (Hint: What does du equal in terms of θ ?)

6. Use the trig identity $\sin^2(\theta) + \cos^2(\theta) = 1$ or $\tan^2(\theta) + 1 = \sec^2(\theta)$ to simplify the denominator. (This kind of simplification is the goal of trig substitution.)

7. The resulting trig integral is:

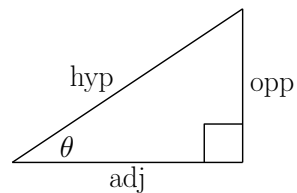
$$\int \frac{(2 \sec(\theta) + 3)(2 \sec(\theta) \tan(\theta))}{32 \tan^5(\theta)} d\theta.$$

(If what you have so far is not equivalent to this, go back and check your work and/or ask your TA for help.) Multiply out the numerator and simplify to obtain the two integrals:

$$\frac{1}{8} \int \frac{\sec^2(\theta)}{\tan^4(\theta)} d\theta + \frac{3}{16} \int \frac{\sec(\theta)}{\tan^4(\theta)} d\theta.$$

8. Evaluate these trig integrals. You will need to use more substitutions (use a new variable, like w) and trig identities. If you don't see what substitution to use, you may try converting the secants and tangents to sines and cosines. The answer in terms of θ is $\frac{-1}{24} \cot^3(\theta) - \frac{1}{16} \csc^3(\theta) + \frac{3}{16} \csc(\theta) + C$.

9. Convert back to the variable u by using our chosen trig substitution and the definition of the trig functions in terms of a right triangle.



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sec \theta =$$

$$\cos \theta =$$

$$\csc \theta =$$

$$\tan \theta =$$

$$\cot \theta =$$

10. Write your answer in terms of the variable x .